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Characterization and Functional Evaluation of a Fiber-Optic Instrumented Impedance-Controlled Upper-Limb Exoskeleton

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ABSTRACT

The use of active exoskeletons for musculoskeletal disorder (MSD) prevention is increasing in industrial environments. Wearable sensing technologies enable closed-loop control of these devices. However, magnetic disturbances and sensor fixation issues can compromise measurement reliability and affect exoskeleton behavior. Polymer optical fiber (POF) sensors offer intrinsic immunity to electromagnetic interference and mechanical compliance, making them a promising alternative for wearable kinematic sensing. Nevertheless, their integration within the control loop of active exoskeletons remains largely unexplored. This study investigates the impact of POF sensing modality integrated within the impedance control loop of an upper-limb exoskeleton, focusing on control performance and system dynamics. A two-stage experimental protocol was conducted under an approved ethical protocol. First, a test-bench evaluation showed that POF sensing modality achieves comparable rise time and mean absolute error to IMU sensing modality, though it resulted in a 0.565 s increase in settling time. Second, participants performed standardized lifting tasks under assisted (Exo-POF and Exo-IMU) and unassisted (No-Assist) conditions. Kinematic and electromyographic (EMG) data showed a 16–23% reduction in muscle activation while preserving range of motion. These findings demonstrate that POF sensing modality significantly influences system dynamics and human–robot interaction. Subsequently, the proof of concept suggests the feasibility of the proposed approach during lifting, handling, and lowering tasks, opening the way for future validation under representative industrial environments.

Introduction

Wearable robots physically coupled to the human body to increase strength, endurance, and functional capabilities across industrial and rehabilitation tasks^{1,2}. Their operation relies on closed-loop control strategies to regulate actuator torque while ensuring safe and stable human–robot interaction³. In particular, impedance and admittance control schemes are widely adopted to modulate interaction forces based on estimated system states⁴. Consequently, accurate joint kinematic estimation becomes a critical component of wearable robot performance, directly influencing control stability and interaction behavior in impedance-controlled systems.

Active upper-limb exoskeletons have gained significant attention in industrial and rehabilitation environments due to their potential to assist repetitive and physically demanding tasks^{5–7}. In these systems, impedance control is commonly used to regulate interaction torque based on measured joint position and velocity. Because control torque is directly computed from estimated kinematic states⁸, sensing characteristics become embedded within the closed-loop system, directly influencing its dynamic behavior. However, prior studies have predominantly addressed this dependency from an estimation accuracy perspective, without explicitly analyzing its implications on closed-loop control behavior.

Multiple sensing technologies have been reported for joint kinematic estimation in wearable robotics. Inertial Measurement Units (IMUs) are widely adopted due to their portability and ease of real-time deployment⁹, while incremental encoders are frequently integrated into rigid exoskeleton structures for precise angular measurements. These sensing modalities enable the implementation of different control strategies within Physical Human–Robot Interaction (pHRI) frameworks¹⁰, supporting the regulation of exoskeleton actuation across various application scenarios. In addition, photonic sensing approaches based on polymer optical fibers (POFs) have emerged as conformable alternatives for kinematic estimation¹¹.

Compared to conventional sensing technologies, POF-based sensors offer advantages such as high flexibility, low fabrication cost, and immunity to electromagnetic interference, making them particularly suitable for wearable robotic applications^{12,13}. Prior studies have demonstrated POF-based solutions for joint angle monitoring¹⁴, rotational estimation using braided fiber

configurations^{15,16}, and curvature-based sensing mechanisms¹⁷. However, most of these works focus on standalone sensing performance, with limited consideration of POF-based sensors as feedback signals within closed-loop control systems. Consequently, the influence of POF sensing characteristics including viscoelastic and frequency-dependent behavior on closed-loop dynamic performance in wearable robotic systems remains insufficiently understood. This scenario represents an open challenge for their effective integration as kinematic feedback elements in physical human–robot interaction control strategies.

In this context, recent advances in wearable robotics have promoted the integration of novel sensing technologies into control frameworks for pHRI. Emerging approaches include triboelectric-based sensors for motion tracking¹⁸, soft stretchable capacitive sensors combined with machine learning¹⁹, bioelectronic sensors for electromyographic signal acquisition²⁰, and flexible piezoelectric architectures for gait phase detection²¹. These developments reflect a growing trend toward embedding sensing systems within wearable devices to support real-time estimation and control. However, these approaches are predominantly based on electronic sensing technologies, and the integration of photonic sensing solutions such as POF sensors within closed-loop control architectures for wearable exoskeletons remains largely unexplored, despite their potential advantages for wearable applications.

Motivated by these limitations, the present study investigates the integration of the POF sensing modality within an impedance control loop applied to active exoskeleton actuation. To achieve this objective, an upper-limb exoskeleton for industrial applications instrumented with a POF sensing system is employed, where the joint angle is used as the primary feedback signal for the impedance controller. A human-in-the-loop experimental study is conducted to assess system performance under real operating conditions, analyzing both kinematic and muscle activity responses. To further characterize system behavior, a controlled test-bench experiment is performed to quantify the influence of sensing characteristics on closed-loop dynamics. The results demonstrate not only the feasibility but also the quantifiable impact of integrating POF sensing modality within closed-loop control architectures, supported by both controlled test-bench analysis and systematic human-in-the-loop evaluation.

Methods

A test-bench was developed to evaluate the POF-based angle sensor under controlled conditions. In parallel, an upper-limb exoskeleton was developed to integrate the proposed sensing approach and to provide assistance during load lifting, handling, and lowering tasks. In this context, this section provides a detailed description of the exoskeleton structure, sensing architecture, test-bench setup, control architecture, and experimental protocol.

Exoskeleton Structure

The proposed upper-limb exoskeleton was designed to assist elbow flexion movements during the lifting, carrying and lowering load tasks (see Figure 1(c)). The exoskeleton structure incorporates a soft human–robot interface based on a modified full-body fall arrest harness, which provides load distribution and mechanical anchoring for the rigid back module of the system. This configuration enables stable attachment of the actuation system while distributing interaction forces across the user's torso. In addition, a lumbar support brace was integrated at the abdominal level to enhance mechanical coupling across the user and the exoskeleton, reducing relative displacement during actuation. To further improve structural stability, reinforced quick-release load-bearing buckles were incorporated into the soft interface to prevent exoskeleton displacement and ensure consistent fixation during dynamic movements.

A rigid 3D-printed interface fabricated in Poly Lactic Acid (PLA) was employed to couple the actuation system to the user's torso, allowing trunk mobility. First, the module called the exospine provides approximately 8° of mobility per segment in the frontal plane (abduction–adduction) and up to 51° in the sagittal plane, allowing natural trunk flexion while serving as the primary load-transmission structure. The electric actuation module is integrated within this lumbar module²². Second, a rigid module extends along the thoracic–cervical region and provides a sagittal Range Of Motion (ROM) of 20°. This segment functions as a passive structural guide for the transmission cable while avoiding compressive loading on the shoulder region. Finally, a 1-DoF passive joint with a 24° of ROM in the coronal plane is integrated at the distal end of the thoracic–cervical 3D-printed interface. This structure enables proper routing of the actuation cable toward the forearm 3D-printed interface²³.

The exoskeleton structure provides one active Degree of Freedom (DoF) corresponding to elbow flexion–extension. The actuation system proposed for this purpose is a cable-driven system based on a DC electric motor, Dynamixel MX-106T (ROBOTIS, USA), and a 2 mm polypropylene nautical rope used as the transmission cable. This material was selected due to its flexibility and improved spooling behavior under repeated winding cycles compared to low-cost steel cables, which exhibited permanent deformation during preliminary tests. The actuation system is routed from the exospine along the thoracic module and connected directly with a PLA-based forearm interface attached to the user. As presented in Figure 1(c), no rigid mechanical joint is aligned with the anatomical elbow center of rotation. Instead, assistive torque is generated to the forearm via the cable interface, enabling compliant assistance without constraining natural movements in the elbow joint.

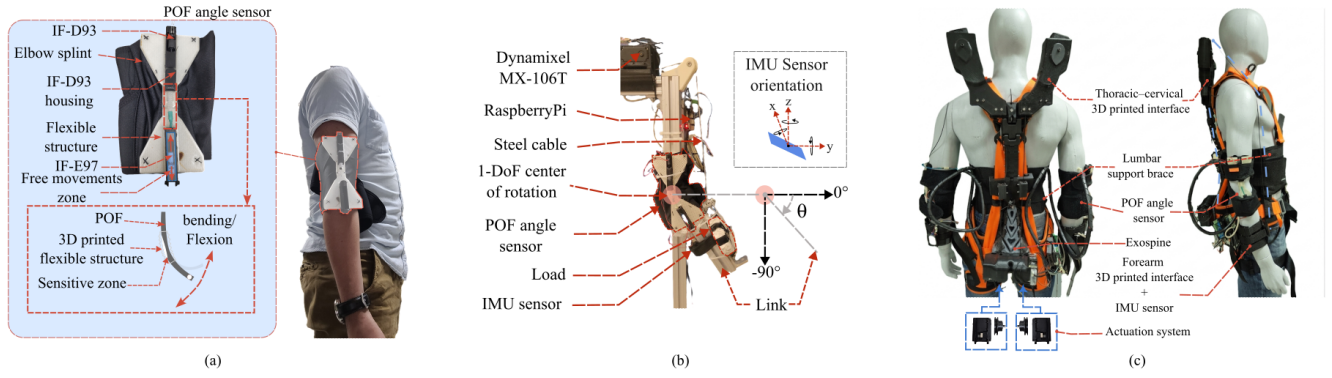


Figure 1

Sensing architecture

The upper-limb exoskeleton was instrumented with both electronic and POF sensing modalities. First, two low-cost 9-DoF Inertial Measurement Units (IMUs) (BNO055, Adafruit, USA), operating at 90 Hz, were mounted on the distal segment of the forearm to capture sagittal-plane orientation during repetitive load-handling tasks. The roll component of the estimated orientation was used as the feedback variable in the impedance control loop under the IMU sensing modality condition.

Second, two POF-based angle sensors that make up the POF sensing modality, built from polymethyl methacrylate (PMMA) POF (SH-4001, Mitsubishi Chemical Corporation), were implemented for elbow angle estimation (see Figure 1(a)). The sensing principle relies on curvature-induced optical attenuation, where fiber bending produces measurable variations in light intensity. To enhance sensitivity, the cladding was modified and a portion of the core was removed to create a localized sensitive region, following²⁴. This region was enclosed within a Fila-Flex 3D-printed tube to reduce mechanical stress during flexion movement.

The sensor was integrated into a textile elbow interface to enable compliant attachment while avoiding rigid coupling to the joint. The optical components consisted of an IF-E97 high-output red Light Emitting Diode (LED) and an IF-D93 photodarlington detector, both housed in rigid PLA-based structures. The housing was designed to securely fix the detector while allowing controlled motion of the emitter along a single axis, thereby reducing mechanical stress in the sensitive region during joint movement. A flexible 3D-printed support, fabricated in Fila-Flex material, was used to couple the sensor to the textile interface while preserving joint mobility. This structure ensures proper alignment of the sensitive region with the elbow rotation axis and prevents undesired torsional deformation of the fiber (Figure 1(a)).

Due to the viscoelastic behavior of the polymer, distinct calibration models were obtained for ascending and descending movements¹¹. Accordingly, cubic polynomial models were fitted to map light intensity variations to angular position for each sensor (Eq. (1)). For the left sensor, RMSE values of 5.55° and 2.61° were obtained for ascending and descending movements, respectively. For the right sensor, RMSE values of 1.12° and 0.59° were obtained.

$$\begin{aligned}
 \theta_{L,asc} &= 2.484 \times 10^{-6} X^3 - 0.047 X^2 + 301.713 X - 6.403 \times 10^5 \\
 \theta_{L,desc} &= 1.664 \times 10^{-6} X^3 - 0.031 X^2 + 202.372 X - 4.299 \times 10^5 \\
 \theta_{R,asc} &= 3.502 \times 10^{-7} X^3 - 0.007 X^2 + 50.066 X - 1.152 \times 10^5 \\
 \theta_{R,desc} &= 3.801 \times 10^{-7} X^3 - 0.007 X^2 + 53.972 X - 1.238 \times 10^5.
 \end{aligned} \tag{1}$$

Where $\theta_{L,asc}$ and $\theta_{L,desc}$ denote the estimated left elbow joint angle during ascending and descending movements, respectively, while $\theta_{R,asc}$ and $\theta_{R,desc}$ correspond to the right elbow joint angle under the same conditions. The variable X represents the output of the IF-D93 module, which is acquired using a 16-bit ADS1115 analog-to-digital converter (ADC) (Texas Instruments). The calibrated POF signal was directly used as the feedback variable within the impedance control loop under the POF sensing modality condition.

Test-bench

As illustrated in Figure 1(b), the test-bench consists of an aluminum-frame structure replicating a one DoF joint operating in the sagittal plane. The arm comprises two 0.255 m links coupled through a 3D-printed connector, with a 1.6 kg load mounted at the distal end to emulate the inertial characteristics of the forearm. The cable-driven actuation system described in the "Exoskeleton structure" subsection was employed to drive the rotational arm, ensuring mechanical consistency with the actuation system.

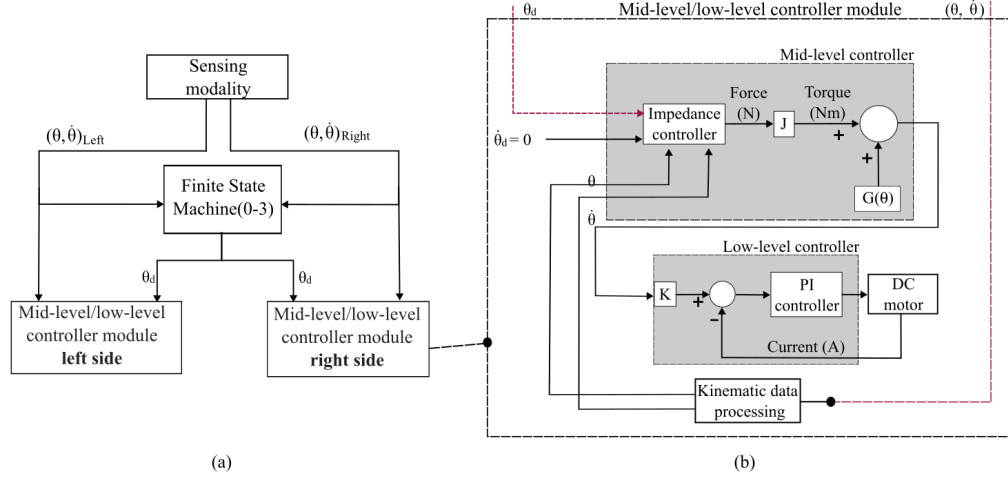


Figure 2

Control architecture

An impedance controller is proposed for the test according to the pHRI methods applied in robots that execute collaborative human tasks³. It is presented as a hierarchical controller where a low-level controller and a mid-level controller are implemented.

Low-level controller: A PI current controller was implemented to regulate the DC motor current and ensure accurate torque values in the cable-driven actuation system (see Figure 2(b)). The controller is defined as:

$$I = K_p(I_d - I) + K_i \int (I_d - I)dt, \quad (2)$$

where I_d represents the desired current generated by the impedance controller and I is the measured motor current obtained through the Hall sensor. The gains for the low-level control were tuned using the Ziegler–Nichols method, where $K_p = 2.94$ and $K_i = 8.66$. Motor torque was estimated from current using an experimentally identified fourth-order polynomial model:

$$T = 0.0523I^4 - 0.4696I^3 + 1.2055I^2 + 0.7472I - 0.0386. \quad (3)$$

Mid-level controller: At the mid-level, an impedance controller was implemented to regulate the interaction torque based on angular position and velocity feedback. The controller models a virtual mass–spring–damper system and is expressed as shown below.

$$\tau = [K_p(x_d - x) - K_d\dot{x}]J_b + G_c(\theta), \quad (4)$$

where θ represents the elbow joint angle obtained from the sensing modality, θ_d is the desired angular position generated by the high-level controller, J_b is the transmission Jacobian, and $G_c(\theta)$ represents gravity compensation. The impedance parameters were selected to achieve the following desired closed-loop characteristics: (i) a damping ratio of $\xi = 0.9$ was selected to approximate a critically damped response, ensuring rapid convergence with minimal oscillatory behavior; (ii) a peak time $t_p = 0.8$ s was selected to match the characteristic duration of voluntary elbow flexion–extension movements reported in the literature (approximately 300–800 ms)²⁵, allowing the controller to respond within the natural time scale of human motion; and a settling time $t_s = 1.1$ s to ensure that the transient response decayed shortly after the movement peak, providing a relatively short stabilization period. Using $w_n^2 = \frac{K_p}{M}$ and $\xi = \frac{K_d}{2w_n M}$, with $M = 1.6$ kg, the controller gains were defined for the equivalent mass–spring–damper system. The control law was implemented in the task space. The angular error $(\theta_d - \theta)$ was first mapped to linear displacement through the transmission mechanism. The resulting position error $(x_d - x)$ and velocity term were then used to compute the interaction force by applying the elasticity constant and damping constant (K_p , K_d). Subsequently, the transmission Jacobian was used to map the force to joint torque. Finally, gravity compensation was applied, where $G_c(\theta) = Mgr\cos(\theta)$, with $g = 9.8$ m/s² corresponding to the gravity acceleration and $r = 0.1$ m.

High-level controller: The high-level control is in charge of regulating the exoskeleton actuation by modifying the desired angular position (θ_d) for the impedance control. Therefore, a four-state Finite State Machine (FSM) was employed to adjust the

exoskeleton actuation based on the measured elbow angular position (θ) and angular velocity ($\dot{\theta}$) for both the right and left sides (See Figure 2(a)). The FSM was divided in four phases: No Actuation; Load Lifting; Carrying Load; Lowering Load.

In the No Actuation Phase, the assistive torque is equal to zero ($T = 0$ Nm). This phase allows the free movement of the user's joints. The transition for the load lifting state is generated when $\dot{\theta} > 0.25$ rad/s for both sides to detect voluntary flexion onset while avoiding false triggering due to sensor noise or minor postural fluctuations. The value was determined during pilot trials to exceed baseline noise levels of POF sensing modality.

The Load Lifting Phase assigns $\theta_d = 0^\circ$ for the right and left sides of the actuation system to provide assistive support during the elbow flexion. The Carrying Load phase maintains $\theta_d = 0^\circ$. Finally, the Lowering Load phase is activated when $\theta \geq 7^\circ$, to allow the extension movement. The angular threshold was empirically determined through preliminary user trials involving lifting, carrying, and lowering tasks. The selected value was intended to prevent unintended activation of the Lowering Load state and to ensure that the state transition occurred at the appropriate stage of the lowering movement. No Actuation phase is activated when $\theta_d \geq -60^\circ$. This angular threshold was empirically determined through preliminary user trials by analyzing the time intervals during which the participant naturally maintained the elbow in an extended position. Once this angular threshold is reached, the impedance controller is deactivated, allowing unrestricted user movement, and the system transitions back to Phase 0. The cable-driven actuation system does not apply additional cable tension. From a system perspective, this angular bound corresponds to the mechanical and ergonomic extension limit of the device and prevents loss of cable tension in the transmission mechanism.

A Raspberry Pi 4 running the Buster operating system was used to acquire sensor data, process signals, and generate the estimated torque commands for actuating the upper-limb exoskeleton. The control architecture was implemented using the Robot Operating System (ROS) Melodic Version, integrating the sensing interfaces, actuation system, and hierarchical control structure. The ADS1115 ADC and the 9-DoF IMU sensors were interfaced with the Raspberry Pi 4 via the I²C communication protocol.

Test protocol

The experimental evaluation was structured in two stages to isolate the effect of sensing modality on the closed-loop dynamics of the impedance-controlled system. The first stage consisted of a test-bench validation, where both sensing modalities (IMU and POF) were used as alternative feedback signals for the impedance controller under mechanically constrained conditions. The second stage involved a preliminary human-in-the-loop validation using the upper-limb exoskeleton, where three experimental conditions were evaluated: No-Assist, Exo-IMU, and Exo-POF. Participants performed repetitive, lifting, carrying, and lowering load tasks to evaluate the performance of the exoskeleton during functional task execution.

Test-bench validation

Both sensing modalities were integrated into the test-bench. The POF sensor was mounted at the center of rotation, while the IMU was positioned at the distal segment of the arm, consistent with the wearable configuration (See Figure 1(b)). The hierarchical controller, including the low-level control and the mid-level impedance controller described in the "Control architecture" subsection, was applied. To evaluate sensing-modality-dependent closed-loop response, each sensing modality was independently used to provide the angular position measurement (θ) and angular velocity ($\dot{\theta}$) to the impedance controller. Two excitation trajectories were applied as reference input (θ_d):

- **Step input:** alternating across -60° and 0° for 9 repetitions.
- **Stepwise sine signal:** with a constant peak-to-peak amplitude of 70° and progressively increasing 0.125 Hz, 0.166 Hz, 0.250 Hz, 0.5 Hz, and 1 Hz.

Each signal was repeated five times per sensing modality. During testing, both IMU and POF measurements were continuously recorded. While one modality was selected as the active feedback signal for closed-loop control, the alternative modality remained enabled for monitoring and comparative analysis.

For the step input experiments, transient performance metrics were extracted separately for flexion and extension movements. The rise time (t_r), peak time (t_p), settling time (t_s), and maximum angular velocity were calculated for statistical analysis. The Mean Absolute Error (MAE) was computed using the IMU signal as a common kinematic reference for both sensing modalities, enabling direct comparison of closed-loop responses. Each transition was treated as an individual transient response, enabling quantitative comparison across sensing modalities. For statistical analysis, the normality of the variables was assessed using the Lilliefors test. Since normality assumptions were not consistently satisfied, statistical comparisons across sensing modalities were performed using the Wilcoxon signed-rank test with a significance level of $p < 0.05$.

For the stepwise sinusoidal excitation, the phase lag was estimated from the averaged system response using the IMU measurement as a unified kinematic reference. This approach allowed the dynamic effects of the feedback modality to be evaluated under a consistent observation framework. The frequency-dependent closed-loop behavior was characterized through

phase lag estimation at each excitation period and by identifying the maximum operational frequency. The sinusoidal excitation analysis was evaluated descriptively to assess dynamic limitations associated with each sensing modality.

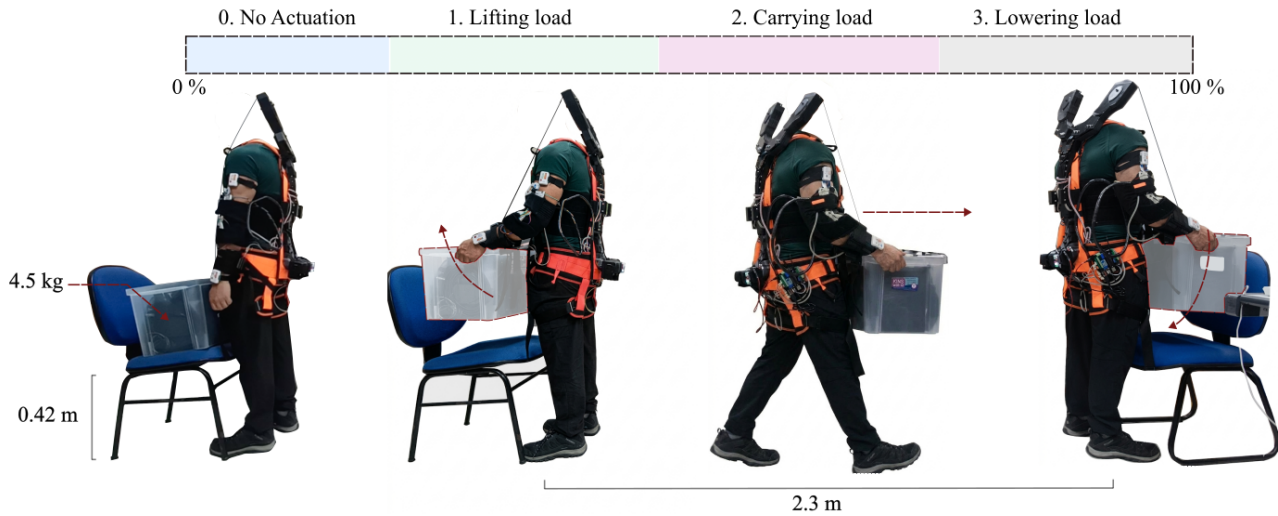


Figure 3

Human-in-the-Loop validation

Healthy participants performed a standardized lifting and carrying task in a single experimental session. The study was approved by the College Research Ethics Committee (CATE), University of the West of England, Bristol, UK (Reference No. 13312133). The methods were performed in accordance with the relevant guidelines and regulations, including the Declaration of Helsinki. All participants provided written informed consent prior to participation. The participants underwent a familiarization period of approximately 20–30 minutes to become accustomed to the device and task execution.

The task consisted of lifting a 4.5 kg load positioned inside a box placed on a platform at a height of 0.424 m. Participants stood in front of the box, grasped it, and performed elbow flexion to lift the load. The box was then transported horizontally over a distance of 2.30 m and subsequently returned to the initial surface through controlled elbow extension (see Figure 3). The movement was self-paced, with no externally imposed timing control. The task was executed under three experimental conditions:

- **Exo-POF**, in which the exoskeleton actuation was driven by the POF sensing modality integrated within the control loop;
- **Exo-IMU**, in which the exoskeleton actuation was driven by the IMU sensing modality integrated within the control loop;
- **No-Assist**, in which the task was performed without Assistance.

For each condition, three healthy participants (age: 28 ± 4.72 years, height: 1.723 ± 0.125 m, weight: 78.46 ± 17.72 kg) completed up to 20 repetitions of the full lifting–carrying–lowering cycle. Due to experimental constraints, one participant completed a reduced number of repetitions (14 trials). The order of conditions was fixed across participants because the proposed Exo-POF prototype required complete integration of the POF sensing module into the exoskeleton, including careful placement and verification of the sensor to ensure reliable angle measurements for the impedance controller. As the POF sensing module was not designed as a modular component and remained physically integrated into the exoskeleton throughout the experimental session, these preparation and verification procedures were performed once for each participant before commencing the trials. Once the integrated sensing and control system had been successfully verified, the Exo-POF condition was therefore performed first, as this was the only experimental condition in which the POF sensing module directly provided the feedback signals for the impedance controller. The No-Assist and Exo-IMU conditions were subsequently conducted according to the experimental protocol using the same instrumented exoskeleton configuration, with rest intervals of 5–10 minutes were provided between consecutive conditions. Kinematic measurements were obtained using inertial measurement units positioned on the distal segment of the forearm, consistent with the sensing architecture of the exoskeleton. One IMU sensor was used per arm and mounted on the forearm segment. During the task execution, participants were instructed to

perform the movement primarily through elbow flexion–extension while minimizing shoulder motion. Under this constrained movement condition, the forearm orientation measured by the IMU was used as an approximation of the elbow joint angle. Surface electromyography (EMG) was recorded from the Biceps Brachii (BB) muscle to monitor muscle activity (FreeEMG, BTS Bioengineering). All signals were synchronously recorded throughout task execution.

The EMG signals were processed using a standard preprocessing pipeline implemented in MATLAB. Raw signals were band-pass filtered using a Butterworth filter, followed by full-wave rectification. A smoothed EMG envelope was then obtained, and the processed signals were used to compute RMS and integrated EMG (iEMG) metrics. To account for inter-subject variability, the EMG signals were normalized with respect to the maximum voluntary contraction (MVC) obtained for each participant. MVC values were estimated from three trials per arm, and the maximum mean activation value across trials was used for normalization. All EMG metrics were computed from the normalized signals.

The MVC tests were performed for the BB muscle on both the left and right arms. During the MVC protocol, participants executed elbow flexion against a fixed mechanical constraint to ensure maximal muscle activation. Each participant performed three flexion attempts per arm, and three MVC trials were recorded for analysis. Each trial lasted 5 s, followed by a 15 s rest interval across repetitions. The maximum value obtained from the MVC trials was used to normalize the EMG signals.

Results

This section presents the results of two Stages. Stage I reports sensing-modality-dependent effects under isolated test-bench conditions, analyzing time-domain transient metrics derived from step inputs and frequency-domain dynamic descriptors obtained from stepwise sinusoidal excitation. Subsequently, Stage II presents the results of the human-in-the-loop evaluation of the impedance-controlled upper-limb exoskeleton using the POF sensing modality, assessing its performance in terms of kinematic and muscle activation metrics during the lifting task.

Test-bench validation

Step-response time-domain transient metrics

Table 1(a) presents the time-domain transient metrics obtained from the step excitation for both sensing modalities, evaluated during flexion and extension movements. For the t_r , significant differences were found during both flexion and extension movements ($p < 0.001$). During flexion, the POF sensing modality reduced t_r by 10.8 % compared to IMU. In contrast, during extension, t_r was slightly higher under POF than IMU. For t_p , significant differences were observed across both modalities for flexion and extension movements ($p < 0.001$). The POF sensing modality resulted in longer t_p during both flexion and extension compared to IMU.

For overshoot, significant differences were observed during both modalities for flexion and extension movements ($p < 0.001$). During flexion, POF produced a substantially higher overshoot than IMU sensing corresponding to an approximately 76% increase relative to IMU. In contrast, during extension, POF reduced overshoot relative to IMU sensing, representing an approximately 34% reduction relative to IMU. Maximum angular velocity also exhibited significant differences across modalities in both movement phases ($p < 0.001$). During flexion, POF presented higher peak velocities than IMU sensing (159.672 °/s vs. 85.136 °/s), corresponding to an approximately 88 % increase relative to IMU. In contrast, during extension, angular velocity shows a moderate increase (144.958 °/s vs. 137.097 °/s representing an approximately 6% increase relative to IMU).

The t_s also exhibited significant differences across sensing modalities during flexion and extension movements ($p = 0.014$ and $p < 0.001$, respectively). During flexion, the IMU sensing modality resulted in a shorter compared to the POF sensing modality. The difference was more pronounced during extension, where the IMU sensing modality achieved $t_s = 1.589$ s, whereas the POF sensing modality required $t_s = 2.155$ s to reach steady-state conditions. Finally, the MAE demonstrated significant differences across sensing modalities ($p = 0.012$), with values of 9.997° under the IMU sensing modality and 10.492° under the POF sensing modality. Finally, Figure 4 corroborates the results summarized in Table 1(a) by presenting the mean trajectory obtained from 45 repetitions of the step-response test. The IMU sensing modality exhibits lower overshoot during flexion movements. In contrast, during extension movements the IMU sensing modality produces larger overshoot relative to the POF sensing modality.

Similarly, the angular velocity profiles in Figure 4(c) reflect the differences in maximum angular velocity. During flexion, the POF sensing modality produces higher peak angular velocities, whereas during extension the IMU sensing modality generates larger velocity peaks. Overall, the trajectories confirm the dynamic trends identified in the quantitative metrics.

stepwise sinusoidal-response frequency domain

The descriptive analysis for the stepwise sinusoidal excitation is summarized in Table 1(b), where phase lag values are reported as median (Q1–Q3) for each sensing modality.

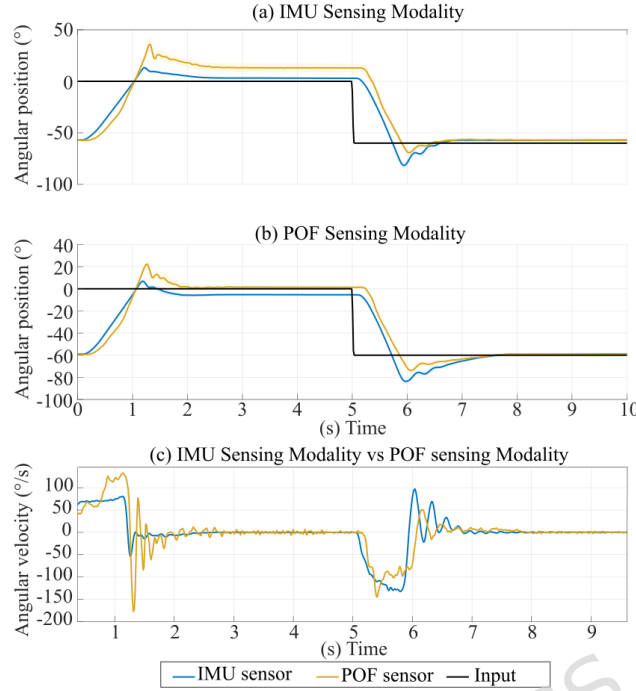


Figure 4

For low and mid frequencies (0.125, 0.166, and 0.250 Hz), the IMU sensing modality exhibited lower phase lag values compared to the POF sensing modality. As the excitation frequency increased (0.5 Hz and 1 Hz), phase lag increased for both modalities. Notably, at the highest tested frequency (1 Hz period), the POF sensing modality exhibited a lower median phase lag compared to the IMU sensing modality.

Human-in-the-loop validation

For Stage II, the EMG metrics evaluated for each user are summarized in Table 2(a). EMG signals from the left and right BB were averaged to obtain a single representative muscle activity profile per participant. The RMS metric reflects the instantaneous muscle activation level of the BB. In this case, both assisted conditions (Exo-POF and Exo-IMU) showed lower RMS values compared to No-Assist condition with reductions ranging from 17% to 20% for Exo-POF condition and 21% for Exo-IMU condition. A particular case was observed for the User 2 where similar RMS values for No-Assist and Exo-IMU conditions were observed. In contrast, the iEMG metric reflects the accumulated muscle activity over time where the Exo-POF and Exo-IMU conditions exhibited higher iEMG values compared with the No-Assist condition. Across users, the increase ranged from 15 % to 84 % for the Exo-POF condition and from 44% to 60% for Exo-IMU. Similarly, task execution time was longer under assisted conditions, where the use of the exoskeleton increased the execution time of the lifting, carrying, and lowering phases.

Figure 5(a) shows the mean EMG profiles obtained from the repetitions for each participant during the lifting, carrying, and lowering load task. The EMG signals were normalized with respect to MVC and the task cycle (0–100%). Across all participants, both assisted conditions generally exhibited lower muscle activity during the carrying phase, where the Exo-POF condition showed reduced variability, as indicated by narrower standard deviation bands. A similar trend was observed under the Exo-IMU condition, which also showed lower muscle activity and similar variability during this phase.

During the lifting phase, Users 1 and 3 showed lower or similar muscle activity under the Exo-POF condition, whereas User 2 exhibited higher activation levels than in the No-Assist condition. On the other hand, lower muscle activity was observed under the Exo-IMU condition for all participants compared with both the No-Assist and Exo-POF conditions. Finally, during the lowering phase, all participants presented higher muscle activity under the Exo-POF condition than under the No-Assist condition. In contrast, Users 1 and 3 showed activation patterns under the Exo-IMU condition that were similar pattern to those observed under the No-Assist condition, whereas User 2 presented higher muscle activity than under the Exo-POF condition.

This behavior is also reflected in the pooled distributions shown in Figure 6(a). The pooled RMS EMG values show a shift toward lower activation levels under the Exo-POF condition, with a more compact distribution indicating reduced variability. The Exo-IMU condition shows a lower activation level than the No-Assist condition. Nevertheless, a comparable variability

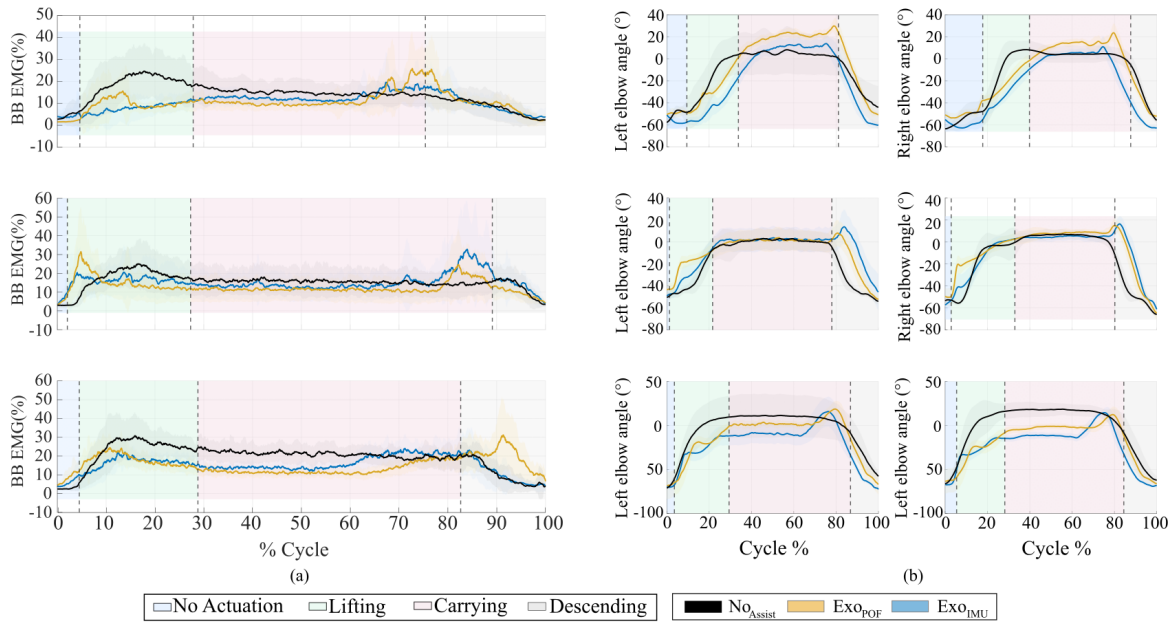


Figure 5

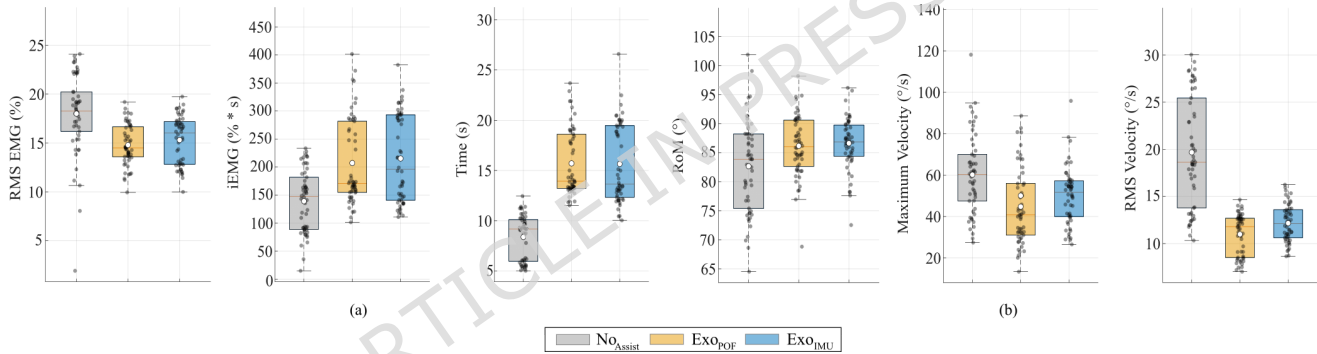


Figure 6

with No-Assist condition and a slightly higher activation levels compared with Exo-POF condition is observed. In contrast, iEMG and task duration showed an increase under the assisted condition (Exo-POF and Exo-IMU conditions). An increase in the variability was observed for the assisted conditions.

Table 2(b) summarizes the elbow kinematic metrics obtained during the No-Assist, Exo-IMU, and Exo-POF conditions. For this preliminary study, the dominant arm of each participant was considered (User 1: left-handed; Users 2 and 3: right-handed). First, the Exo-POF condition showed higher ROM values than the No-Assist condition, with increases of 5.793% and 14.658% for Users 1 and 2, respectively. In contrast, a slight reduction in ROM was observed for User 3. Similarly, the Exo-IMU condition exhibited similar results to the Exo-POF condition, with higher ROM values than those obtained under the No-Assist condition. For maximum angular velocity, lower peak velocities were observed under both assisted conditions than under the No-Assist condition for Users 1 and 3, with reductions of 37.153% and 52.407% for the Exo-POF condition and 27% and 10% for the Exo-IMU condition, respectively. Conversely, User 2 exhibited higher maximum angular velocities under both assisted conditions, with increases of 77% for the Exo-POF condition and 23% for the Exo-IMU condition relative to the No-Assist condition. A similar trend was observed for the RMS angular velocity. Users 1 and 3 exhibited lower RMS angular velocity under both the Exo-POF condition (reductions of 50% and 53%, respectively) and the Exo-IMU condition (reductions of 40% and 43%, respectively). In contrast, User 2 exhibited consistent RMS angular velocity values across the assisted and No-Assist conditions.

Figure 5(b) illustrates the mean elbow angle trajectories for each participant during the lifting–carrying–lowering task, for the left and right arms. The shaded regions represent the standard deviation across repetitions. Across all participants, the

Exo-POF and Exo-IMU conditions generally showed lower trajectory variability compared with the No-Assist condition. These observations agree with the values reported in Table 2(b), where increased ROM values were observed for Users 1 and 2, while similar values were obtained for User 3. Additionally, a peak elbow angle was observed across approximately 70% and 100% of the task cycle, corresponding to the lowering load phase.

Figure 6(b) presents the global effects on the kinematic metrics under each condition, complementing the human-in-the-loop analysis. The pooled distributions show an increase in ROM under the Exo-POF condition (4.23%) and the Exo-IMU condition (4.78%). In contrast, both maximum angular velocity and RMS angular velocity were reduced by 25.49% and 44.15%, respectively, under the Exo-POF condition, and by 16.70% and 38.15%, respectively, under the Exo-IMU condition. The pooled distributions also show reduced variability in ROM and RMS angular velocity under both assisted conditions, whereas maximum angular velocity exhibits a similar pattern across all conditions. Overall, the pooled distributions indicate lower RMS EMG, lower maximum and RMS angular velocity, and slightly higher ROM values under both assisted conditions compared with the No-Assist condition. Reduced variability was also observed across several metrics under the assisted conditions. A preliminary SUS evaluation yielded a mean score of 60.83 across the three participants, indicating marginal usability according to standard SUS interpretation guidelines.

Discussion

Test-bench validation

The t_r results obtained during the step input tests indicate differences across sensing modalities. For flexion movements, the POF sensing modality exhibited a faster dynamic response, with a t_r approximately 0.070 s shorter than that obtained using the IMU sensing modality. In contrast, during extension movements the IMU sensing modality showed a slightly faster response, with a t_r approximately 0.033 s shorter than that observed with the POF configuration. Although the statistical analysis revealed significant differences for both movement directions, the magnitude of these differences is relatively small.

The t_p results also revealed significant differences across sensing modalities. The IMU sensing modality showed a faster response for both flexion and extension movements, with differences of approximately 0.052 s and 0.125 s, respectively. The overshoot percentage further highlighted differences that may influence the dynamic behavior of the robotic system. Particularly, the POF sensing modality presented overshoot values approximately 17 % higher than those obtained with the IMU sensing modality during flexion movements. The increased overshoot is typically associated with a reduction in the effective damping ratio. Therefore, these results suggest that the POF sensing modality may lead to a slightly less damped closed-loop response during flexion.

In contrast, during extension movements the POF sensing modality exhibited overshoot values approximately 12.6% lower than those obtained with the IMU sensing modality, which may indicate a comparatively higher effective damping behavior of the closed-loop system. For flexion movements, the opposite behavior is observed, where the POF sensing modality exhibits a less damped response, as illustrated in Figure 4(c). Consequently, the IMU sensing modality presents a comparatively more damped transient response for both flexion and extension movements. This closed-loop system response is further supported by the t_s results, where for extension movements the IMU sensing modality reaches steady-state approximately 0.565 s faster than the POF sensing modality.

The MAE provides an overall measure of the deviation across the desired joint position and the joint angle values obtained from the sensing modalities. As observed in Table 1(a), statistically significant differences were found across the sensing modalities, with the IMU sensing modality presenting MAE values approximately 0.495° lower than those obtained with the POF sensing modality. However, the magnitude of this difference is relatively small and is unlikely to significantly affect wearable robotic applications. It is important to note that the MAE values were computed using the joint angle information obtained from the IMU sensor in both sensing configurations. Therefore, although the POF sensing modality may generate a slightly less damped closed-loop response during flexion movements, this behavior does not significantly affect the overall trajectory tracking system accuracy.

The stepwise sinusoidal excitation results presented in Table 1(b) illustrate the phase lag variations obtained for five different signal frequencies applied to both sensing modalities. The IMU sensing modality shows a higher phase lag primarily for the sinusoidal signal with a 1 Hz period. In contrast, the interquartile range (IQR) indicates that the POF sensing modality exhibits greater variability across experimental repetitions. Consequently, the POF sensing modality demonstrates increased phase lag variation as the period of the sinusoidal signal increases. This behavior may be associated with the viscoelastic and time-dependent mechanical properties of POF, which can influence the dynamic response of the sensor^{11,26}.

The analysis performed in Stage I revealed statistically significant differences in the t_r and t_p across sensing modalities. However, the magnitude of these differences is relatively small and is not expected to affect the execution of typical upper-limb tasks. In particular, voluntary elbow flexion–extension movements are commonly reported to occur within time ranges of approximately 300 ms to 800 ms, depending on the execution speed²⁵. Therefore, the differences representing only a small fraction of the overall movement. However, not all transient metrics exhibit the same behavior across sensing modalities.

A particular case is observed in the settling time results across sensing modalities, where a significant difference is obtained for extension movements, with variations of approximately 0.56 s. This difference represents a noticeable change in the transient response of the closed-loop system when compared with the time scales reported for voluntary elbow flexion–extension movements²⁵. The slower settling time observed during extension movements may be associated with hysteresis effects related to the viscoelastic behavior of polymer optical fiber sensors. As reported in²⁷, mechanical stress applied to the POF sensing region can induce relaxation phenomena that influence the dynamic response of the sensor.

Human-in-the-loop validation

The muscle activity results show a reduction in RMS activation for all users during the task when using the exoskeleton. The relative reduction in muscle activity is consistent with values reported in previous studies on repetitive task execution with assistive devices^{28,29}. A particular case was observed for User 2, where the RMS EMG observed under the Exo-IMU condition was slightly higher than that obtained under the No-Assist condition. In this regard, the mean normalized EMG profile revealed a pronounced activation peak during the lowering phase, which likely increased the RMS value computed over the complete task cycle. This peak may be related to the activation of the lowering phase by the FSM, which requires the elbow angle to reach a predefined threshold before initiating this phase.

In contrast, the iEMG metric indicates higher accumulated muscle activity under the Exo-POF and Exo-IMU conditions. This increase is associated with the longer task duration observed when using the exoskeleton, where the task execution time was higher than that of the No-Assist condition. These results suggest that, although the Exo-POF and Exo-IMU conditions showed lower instantaneous muscle activation, they are also associated with longer task execution times. A similar behavior has been reported in the literature, where iEMG is influenced by both EMG amplitude and task duration, since changes in either variable can independently affect the accumulated muscle activity³⁰. In this case, the longer task execution time appears to be one of the main factors contributing to the increase in iEMG. This was observed in the results obtained for User 2, where a reduction in RMS EMG under the Exo-POF condition was accompanied by only a slight increase in task execution time, resulting in a smaller increase in iEMG than that observed for Users 1 and 3, who exhibited larger increases in task execution time. Considering that the primary objective of wearable robotic systems is to reduce muscular demand³¹, additional performance metrics should be considered to provide a more comprehensive evaluation of the device. In particular, as mentioned in³², task execution time is an important metric because it reflects the duration over which the body remains exposed to the external load. In this sense, reduced EMG activity accompanied by a prolonged task execution time may reduce the potential benefits of the exoskeleton. In addition, increased task execution time has also been reported in both passive and active exoskeletons. In the present study, this behavior is likely associated with the dynamic limitations of the actuation system, specifically the maximum actuator velocity and available output torque of the device³².

In addition to the muscle activity analysis, the kinematic parameters showed comparable trends, with similar or slightly higher ROM values under the Exo-POF and Exo-IMU conditions than under the No-Assist condition across participants. These results indicate that neither the Exo-POF nor the Exo-IMU condition restricted elbow motion during the task. This behavior may be explained by the mechanical design of the system, where the cable-driven actuation mechanism is not rigidly aligned with the joint center of rotation, allowing the user to maintain a natural movement trajectory. As reported in previous studies, an effective exoskeleton design should reduce muscular demand while preserving a large ROM³³. However, the FSM proposed for the active exoskeleton controller may have changed the natural elbow movement, as reflected by the slight increase in ROM. This behavior may be related to the threshold-based detection implemented in the FSM to initiate the lowering phase. The slight variations in the angle trajectories across conditions may also be attributed to motor adaptation when interacting with the exoskeleton and its control system^{34,35}. Such adaptations are commonly observed in wearable robotic systems, where users adjust their movement patterns to accommodate the mechanical dynamics and assistance provided by the device.

Some of the differences observed between the kinematic trajectories between the Exo-IMU and Exo-POF conditions may be related to the slight ROM differences observed between the left and right sides. As described in the Methods section, the left and right POF-angle sensors exhibited different RMSE values after calibration. In this sense, the following factors may have been affected. (i) The current elbow joint angle used as feedback relies on the estimation provided by the corresponding POF-angle sensor. Therefore, differences in calibration accuracy may introduce slight differences in the measured angular position, which is used by the virtual spring element of the impedance controller to compute the angular position error between the desired and current elbow joint angles. Consequently, slight differences in the assistive torque generated by each impedance controller may occur between both sides. (ii) A similar situation is observed for the elbow joint angular velocity, which was computed as the first time derivative of the estimated elbow joint angle. In this sense, the differences in the RMSE of each POF-angle sensor may lead to differences in the estimated angular velocity for each side. As a result, the virtual damper element of the impedance controller may generate slight differences in the assistive torque between the left and right sides. Finally, (iii) although the impedance control loops for the left and right arms operate independently, the high-level bilateral coordination implemented through the proposed finite-state machine (FSM) relies directly on the angular position estimated by

each POF-angle sensor. In particular, the transition to the Lowering Load phase requires that both estimated elbow joint angles exceed the predefined threshold ($\theta \geq 7^\circ$). Therefore, small calibration differences could slightly modify the timing of the FSM state transition.

Beyond the ROM analysis, User 2 presented higher maximum angular velocity under the Exo-POF condition, followed by the Exo-IMU condition. This behavior may be associated with two factors previously observed in the analysis: (i) the relatively small difference in task execution time across conditions for this participant, and (ii) the smaller reduction in muscle activity obtained for User 2 when using the exoskeleton. As reported in previous studies, increases in movement frequency may degrade the performance of exoskeleton assistance due to phase delays in the control loop³⁶. This interpretation is consistent with the RMS angular velocity results, where User 2 exhibited similar values across conditions, whereas Users 1 and 3 presented lower RMS angular velocity under the Exo-POF condition than under the No-Assist condition.

Additionally, reduced trajectory variability was observed under both the Exo-POF and Exo-IMU conditions. A reduction in movement variability has also been reported in studies involving wearable robotic devices³⁷. However, the mechanisms underlying this behavior remain an open research question and require further investigation to fully understand their implications for human-exoskeleton interaction.

Considering that the experimental conditions were performed in a fixed order across participants, learning, adaptation, fatigue, or familiarization effects may have influenced the observed outcomes. However, the experimental protocol included measures to account for some of these factors. Specifically, rest periods of 5–10 minutes were provided between experimental conditions, which are consistent with previous recommendations for muscle recovery³⁸. In addition, a familiarization session was conducted prior to data collection as part of the experimental protocol. Nevertheless, residual learning, adaptation, fatigue, or familiarization effects during the experimental trials cannot be completely ruled out. Furthermore, the human-in-the-loop evaluation was conducted under controlled laboratory conditions rather than representative industrial environments. It should also be noted that this study was conceived as a preliminary evaluation aimed at assessing the feasibility and functionality of the proposed Exo-POF sensing approach rather than providing an exhaustive comparative assessment. Accordingly, future developments will focus on simplifying the integration of the POF sensing module into the exoskeleton, improving the practicality of the proposed system. In addition, future studies involving larger populations of industrial workers under representative industrial conditions and fully randomized experimental designs are needed to enable a more robust statistical assessment and to further validate the findings of the present study.

The marginal SUS score obtained in this study may be associated with the increased task execution time observed under the assisted condition, may also reflect the initial user adaptation to the exoskeleton and its control dynamics. It is important to note that the usability evaluation was conducted with a limited number of participants, and further studies with larger cohorts and extended familiarization periods are required to better assess user experience.

Conclusion

Closed-loop impedance control of active exoskeletons for industrial applications relies on the accurate acquisition of human kinematic parameters. However, wearable sensing systems used for this purpose may be affected by external factors present in industrial environments, potentially influencing the dynamic behavior of the exoskeleton. This work presented the development and systematic experimental evaluation of an upper-limb exoskeleton for industrial applications instrumented with a polymer optical fiber (POF) sensor for joint angle estimation. The experimental evaluation yielded the following main findings. First, the controlled test-bench experiment demonstrated that the POF sensing modality exhibited significant differences in transient and frequency responses, including variations in settling time, overshoot, and phase lag, despite comparable tracking accuracy. Second, the human-in-the-loop assessment applied for No-Assist, Exo-POF, and Exo-IMU conditions provided a descriptive assessment in terms of BB muscle activity and kinematic parameters of the forearm link. As a result, the preliminary human-in-the-loop validation provided initial evidence of reduced BB activation and more regulated forearm movement patterns while preserving the ROM with slightly increases due to the FSM applied for the active exoskeleton actuation. These findings indicate the influence of sensing characteristics on closed-loop system dynamics. In addition, the proof of concept of POF sensing modality integrated into an upper-limb exoskeleton showed a feasible alternative for wearable sensing in impedance-controlled exoskeletons. For future works, the application of a long-term study with a larger study population evaluated in representative industrial environments and a different algorithm for motion intention detection can further validate the benefits of the employment of POF sensing modality for monitoring and control wearable robotic devices or industrial applications.

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Author contributions

L.J.A.M. conceived the study, designed and conducted the experiments, performed the data analysis, and wrote the manuscript. P.S. contributed to the development and execution of the experimental protocol. M.E.V.S., C.A.C., and C.A.R.D. supervised the research and provided guidance throughout the project. All authors reviewed and approved the final manuscript.

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Competing interest

The authors declare no competing interests.

Data availability

The datasets generated during the current study are publicly available in the Figshare repository at: <https://doi.org/10.6084/m9.figshare.31864330>.

Ethics declarations

The study was approved by the College Research Ethics Committee (CATE), University of the West of England, Bristol, UK (Reference No. 13312133). All participants provided written informed consent prior to participation.

Figures legends

Figure 1. Upper-limb exoskeleton used for the lifting–carrying–lowering task. (a) POF-based angle sensor developed using textile materials and flexible 3D-printed interfaces for elbow joint angle estimation. (b) Test bench adapted to assess the effect of using the POF angle sensor as feedback for an impedance controller. (c) Mechanical structure consisting of a rigid PLA frame mounted on the user’s back, with textile components for body attachment..

Figure 2. Control architecture of the upper-limb exoskeleton actuation system. (a) FSM serving as the high-level controller for task phase identification, where the sensing modality module corresponds to either the IMU sensing modality or the POF sensing modality, selected as the direct feedback for the impedance controller. (b) Low-level control module integrating an impedance controller and a current controller to generate assistive torques during the lifting, carrying, and lowering phases.

Figure 3. One complete lifting–carrying–lowering cycle performed during the human-in-the-loop validation protocol. The participant lifts a 4.5 kg load, transports it over 2.3 m, and places it on the opposite side. The protocol was conducted under both No-Assist and Exo-POF conditions. Image contrast and resolution were enhanced using non-generative AI-based tools for clarity, without altering the underlying content.

Figure 4. Step response of the impedance-controlled test-bench under different sensing modalities. A step input was applied as the excitation signal to evaluate the system dynamics. (a) Angular position response when the controller operates using the IMU sensing modality. (b) Angular position response when the controller operates using the POF sensing modality. (c) Corresponding angular velocity response of the system for both sensing modalities (IMU and POF sensors).

Figure 5. EMG and kinematic metrics time-normalized to the task cycle (0–100%) during the lifting–carrying–lowering task. Each row corresponds to Users 1–3 for No-Assist, Exo-IMU, and Exo-POF conditions: (a) Normalized Biceps Brachii (BB) EMG activity for each user; solid lines represent the mean profile across repetitions and shaded regions indicate the standard deviation; (b) Mean elbow angle trajectories (degrees). Left and right columns represent the left and right elbow angles, respectively, with shaded regions indicating the standard deviation.

Figure 6. Distribution of muscle activity and kinematic metrics pooled across participants for No-Assist, Exo-IMU, and Exo-POF conditions. Each box represents the interquartile range (IQR) and each point corresponds to an individual repetition. Significant differences between conditions were assessed using a paired t-test on participant-level mean values. (a) Muscle activity metrics: RMS EMG, iEMG, and task duration. (b) Kinematic metrics: ROM, maximum velocity, and RMS velocity.

Tables

Table 1. Test-bench validation for transient performance metrics under IMU and POF sensing modalities; (a) step response metrics (median [Q1–Q3]); (b) Phase angle under stepwise-sine excitation (median Q1–Q3) in degrees.

(a)				
Metric	Phase	IMU sensing modality	POF sensing modality	p-value
Rise time (s)	Flexion	0.646 (0.631–0.646)	0.576 (0.576–0.576)	$p < 0.001$
	Extension	0.440 (0.440–0.455)	0.473 (0.473–0.487)	$p < 0.001$
Peak time (s)	Flexion	1.218 (1.203–1.233)	1.270 (1.240–1.285)	$p < 0.001$
	Extension	0.944 (0.929–0.959)	1.069 (1.039–1.098)	$p < 0.001$
Overshoot (%)	Flexion	22.830 (21.391–23.940)	40.148 (38.571–42.681)	$p < 0.001$
	Extension	37.388 (36.268–38.808)	24.785 (22.430–26.411)	$p < 0.001$
Max angular velocity (°/s)	Flexion	85.141 (83.193–92.819)	159.683 (152.464–166.673)	$p < 0.001$
	Extension	137.108 (133.613–139.572)	144.958 (139.515–148.327)	$p < 0.001$
Settling time (s)	Flexion	1.8625 (1.6599–2.0090)	1.9285 (1.8319–2.1424)	$p = 0.014$
	Extension	1.5894 (1.5700–1.6170)	2.1545 (2.0726–2.3248)	$p < 0.001$
MAE (°)	—	9.997 (9.831–10.265)	10.492 (10.001–10.856)	$p = 0.012$

(b)		
Period (Hz)	IMU (deg)	POF (deg)
0.125	28.468 (28.159–28.665)	31.618 (31.224–31.871)
0.166	11.102 (11.027–11.477)	13.352 (12.827–14.065)
0.250	20.704 (19.748–20.873)	23.404 (23.179–24.642)
0.5	26.555 (25.543–28.581)	33.306 (29.930–38.256)
1	45.009 (44.784–45.684)	35.106 (29.255–45.007)

Table 2. Physiological and kinematic metrics across No-Assist, Exo-IMU, and Exo-POF conditions for each participant. Values are reported as mean $\pm$ SD. (a) EMG metrics; (b) Elbow kinematic metrics.

(a)				(b)			
Metric	No-Assist	Exo-IMU	Exo-POF	Metric	No-Assist	Exo-IMU	Exo-POF
User 1				User 1			
RMS	15.852 $\pm$ 4.718	12.433 $\pm$ 0.869	13.138 $\pm$ 1.364	ROM	83.402 $\pm$ 6.009	86.466 $\pm$ 4.279	88.234 $\pm$ 4.523
iEMG	83.641 $\pm$ 21.717	134.295 $\pm$ 14.678	154.352 $\pm$ 21.717	Max velocity	1.154 $\pm$ 0.536	0.838 $\pm$ 0.155	0.726 $\pm$ 0.640
Time	5.754 $\pm$ 0.605	12.789 $\pm$ 0.815	13.454 $\pm$ 1.193	RMS velocity	0.459 $\pm$ 0.045	0.247 $\pm$ 0.020	0.220 $\pm$ 0.015
User 2				User 2			
RMS	16.482 $\pm$ 2.193	17.068 $\pm$ 1.661	14.164 $\pm$ 1.235	ROM	74.387 $\pm$ 4.970	83.382 $\pm$ 4.580	85.291 $\pm$ 7.139
iEMG	145.081 $\pm$ 16.287	210.008 $\pm$ 57.338	168.228 $\pm$ 10.457	Max velocity	0.668 $\pm$ 0.115	0.827 $\pm$ 0.266	1.189 $\pm$ 0.228
Time	9.497 $\pm$ 0.931	13.817 $\pm$ 4.216	13.321 $\pm$ 0.536	RMS velocity	0.216 $\pm$ 0.018	0.209 $\pm$ 0.019	0.212 $\pm$ 0.015
User 3				User 3			
RMS	21.203 $\pm$ 2.297	16.949 $\pm$ 0.835	16.894 $\pm$ 1.261	ROM	87.716 $\pm$ 7.009	89.042 $\pm$ 2.914	84.697 $\pm$ 3.861
iEMG	195.347 $\pm$ 25.757	300.094 $\pm$ 16.163	297.489 $\pm$ 44.613	Max velocity	1.101 $\pm$ 0.226	0.944 $\pm$ 0.285	0.524 $\pm$ 0.189
Time	10.243 $\pm$ 0.860	19.810 $\pm$ 1.011	19.622 $\pm$ 1.852	RMS velocity	0.317 $\pm$ 0.031	0.180 $\pm$ 0.016	0.146 $\pm$ 0.015